

Replication Appendix: Historical Trade-War Tariffs and Import Pass-Through

July 2026

1 Data Sources and Construction

1.1 Trade values and quantities

We construct monthly U.S. imports from the U.S. Census Bureau fixed-width merchandise-trade archives. The unit of observation is partner country c , ten-digit Harmonized System product i , and month t . The source fields are general customs value, general CIF value, first quantity, ordinary duty value, and calculated duty value. Each archive is parsed independently, aggregated to the canonical key, reconciled to its source total, and stored as ZSTD-compressed Parquet with source fingerprints.

Let CIF_{ict} denote general CIF value, Q_{ict} first quantity, and D_{ict} calculated duties. The four import outcomes are

$$m_val_{ict} = CIF_{ict}/10^6, \quad m_q1_{ict} = Q_{ict}/10^6, \quad (1)$$

$$m_p_{ict} = CIF_{ict}/Q_{ict}, \quad m_pduty_{ict} = (CIF_{ict} + D_{ict})/Q_{ict}. \quad (2)$$

Unit values are defined only for positive quantities. Missing quantities and observed zeros are retained as distinct source states. In particular, duty-inclusive unit value uses observed Census calculated duties; it is not constructed by multiplying price by the independently reconstructed tariff.

1.2 Tariffs

The independently reconstructed bilateral tariff combines the applicable most-favored-nation rate with the trade-war increments:

$$\tau_{ict} = \tau_{it}^{MFN} + \Delta\tau_{ict}^{201} + \Delta\tau_{ict}^{232} + \Delta\tau_{ict}^{301}. \quad (3)$$

The baseline and product schedules are taken from official USITC HTS releases, presidential proclamations, Federal Register annexes, and USTR Section 301 notices. Structured tables are used when available; reviewed PDF-table extractions are used for official annexes without a suitable tabular release. Each component retains its source, rate, legal date, and product/partner scope.

Consistent with the original paper’s appendix, the construction excludes antidumping and countervailing duties, unrelated treaty changes, and quota-contingent increments that apply only once an entry-level threshold is crossed. Monthly partner–HS10 data do not identify the tier used by an individual customs entry. The original authors report that the affected quota-threshold varieties account for approximately \$16 million out of roughly \$300 billion in annual targeted imports.

1.3 Timing conventions

If an increment $\Delta\tau$ takes effect on day d of a D -day month, the locked day-weighted initial-month increment follows days in effect:

$$\Delta\tau_t^{dw} = \Delta\tau \frac{D-d}{D}, \quad \Delta\tau_{t+1}^{remainder} = \Delta\tau \frac{d}{D}. \quad (4)$$

The paper-compatible event clock assigns the legal month when $d \leq 15$ and the following month when $d > 15$. The legal clock instead uses the calendar month containing the effective date. Only the paper-clock reconstruction is compared to the paper; the legal-clock curve is an alternative timing exercise.

Untreated products require an event origin to remain in the event-study controls. Following the Stata program, the date is assigned from the earliest date in the same NAICS4 industry, then NAICS3, then NAICS2. February 2018 is the final fallback. This operation does not change treatment status.

2 Replication and Extension Results

2.1 Event-study specification

The locked event study estimates

$$100 \log y_{ict} = \sum_{k=-5}^6 \beta_k \mathbf{1}\{T_i = 1, event_{it} = k\} + \sum_{k=-5}^6 \gamma_k \mathbf{1}\{event_{it} = k\} + \alpha_{ic} + \delta_{ct} + \eta_{it} + \varepsilon_{ict}, \quad (5)$$

where month -6 is omitted and the right tail is top-coded at $6+$. The fixed effects are product-country, country-month, and product-month. Standard errors are clustered by HS8 and partner country.

2.2 Dynamic specification

Let $x_{ict} = \Delta \log(1 + \tau_{ict})$ and let $\Delta \ell y_{ict}$ be the first difference of the corresponding log outcome. We estimate

$$\Delta \ell y_{ict} = \sum_{k=1}^6 \phi_{-k} F^k x_{ict} + \phi_0 x_{ict} + \sum_{k=1}^K \phi_k L^k x_{ict} + \eta_{it} + \delta_{ct} + \kappa_{cs} + u_{ict}, \quad (6)$$

including the missing-lead and missing-lag indicators used by the original Stata code. Fixed effects are product-month, country-month, and country-NAICS4; clustering is by HS8 and country. Plotted effects are cumulative linear combinations, with standard errors computed from the full covariance matrix. The paper replication sets $K = 6$. A separate extension sets $K = 24$ and does not alter the locked benchmark.

2.3 Validation evidence

The package-only Python estimator produces eight import fits with thirteen horizons each. Across 104 PDF reference points, the maximum absolute difference is 1.009620 log points, below the preregistered 1.10 tolerance.

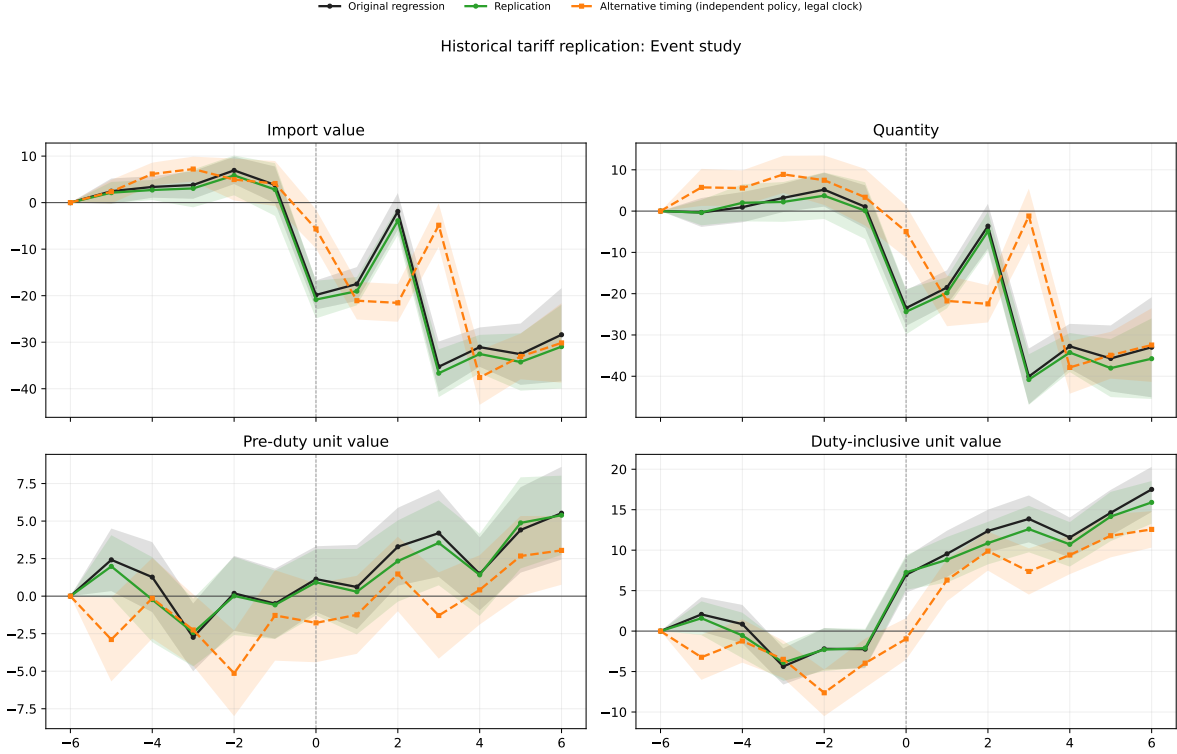


Figure 1: Locked historical event-study replication, $[-6, +6]$.

Table 1 compares the independently reconstructed paper-clock policy with the authors' policy anchor while holding the raw Census outcome sample fixed. Seven of eight point-estimate comparisons satisfy the registered thresholds. Event-study quantity is the disclosed exception.

The accepted decision is therefore *historical methodology accepted with a disclosed quantity exception*. It is not an assertion that the formal eight-of-eight policy gate passed. The corresponding figures distinguish the original package regression, the independent paper-clock replication, and the independent legal-clock alternative.

2.4 Longer horizon

The separate $[-6, 24]$ exercise uses archive-native CIF trade through October 2020. The original package series remains locked to $[-6, 6]$. Because post-April-2019 actions are outside the validated historical tariff ledger, the extension freezes the April 2019 bilateral tariff level. It is therefore a persistence exercise under the locked historical treatment, not an estimate of later tariff actions and not a 2025 event study.

3 The 2025 Tariff Episode

3.1 Archive-native trade panel

The new-episode construction uses the Census fixed-width import archives directly and does not reuse the policy-bearing historical workhorse. The current event panel covers August 2024 through December 2025, or 17 months, and contains 4,949,734 unique partner–HS10–month observations.

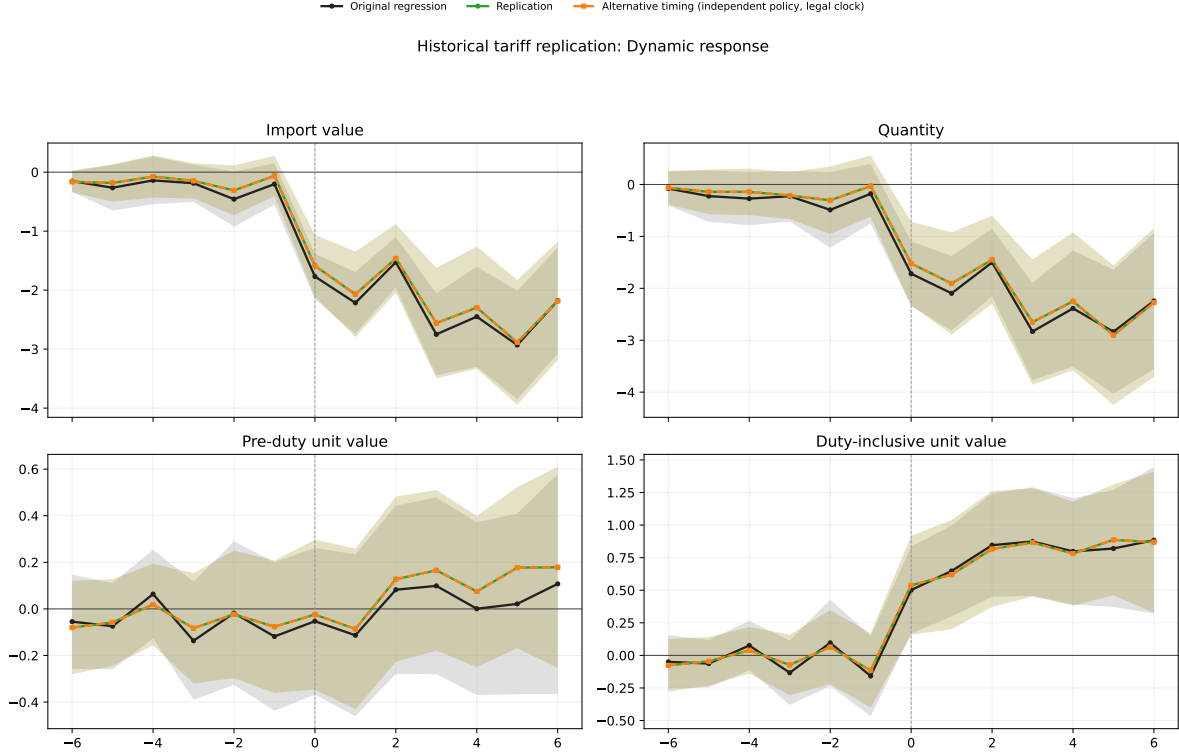


Figure 2: Locked historical dynamic pass-through replication, $[-6, +6]$.

Every monthly CIF total reconciles exactly to its source member, all partitions use ZSTD compression, and no package-policy variable enters the panel.

The February 2025 event therefore has a complete $[-6, +6]$ window and an available long window of $[-6, +10]$. The registered long target remains $[-6, +24]$ and will expand without zero-padding when trade through February 2027 becomes available.

3.2 Policy construction and present gate

The policy source inventory records the China IEEPA action effective February 4, the March increase, reciprocal tariffs and their China escalation and suspension, and the 2025 Section 232 steel, aluminum, and automobile actions. The legal tariff path uses the locked initial-month weight $(D - d)/D$ for an action effective on day d of a D -day month; the event clock assigns dates after the fifteenth to the next month.

Estimation is not yet authorized. The local archive contains 33 official 2025 HTS releases, but the product-level ledger still requires materializing and reviewing versioned reciprocal-tariff exclusions, content-dependent Section 232 rules, automobile USMCA/content treatment, complete partner modifications, and the inherited December 2024 MFN/201/232/301 baseline on the 2025 HS10 universe. Observed calculated duties are retained as an outcome component but are not substituted for the statutory treatment. Accordingly, the following figure hooks remain empty until the policy gate passes.

Table 1: Independent policy substitution on the common raw-outcome sample

Specification	Outcome	Correlation	RMSE	Maximum difference
Event	Value	0.9994	1.0727	1.9403
Event	Quantity	0.9984	1.5650	2.6478
Event	Pre-duty price	0.9703	0.5280	1.4191
Event	Duty-inclusive price	0.9979	0.4691	1.4172
Dynamic	Value	0.9986	0.1203	0.2338
Dynamic	Quantity	0.9970	0.1663	0.2989
Dynamic	Pre-duty price	0.9599	0.0307	0.0595
Dynamic	Duty-inclusive price	0.9978	0.0404	0.0891



Figure 3: Historical event-study persistence exercise, $[-6, +24]$.

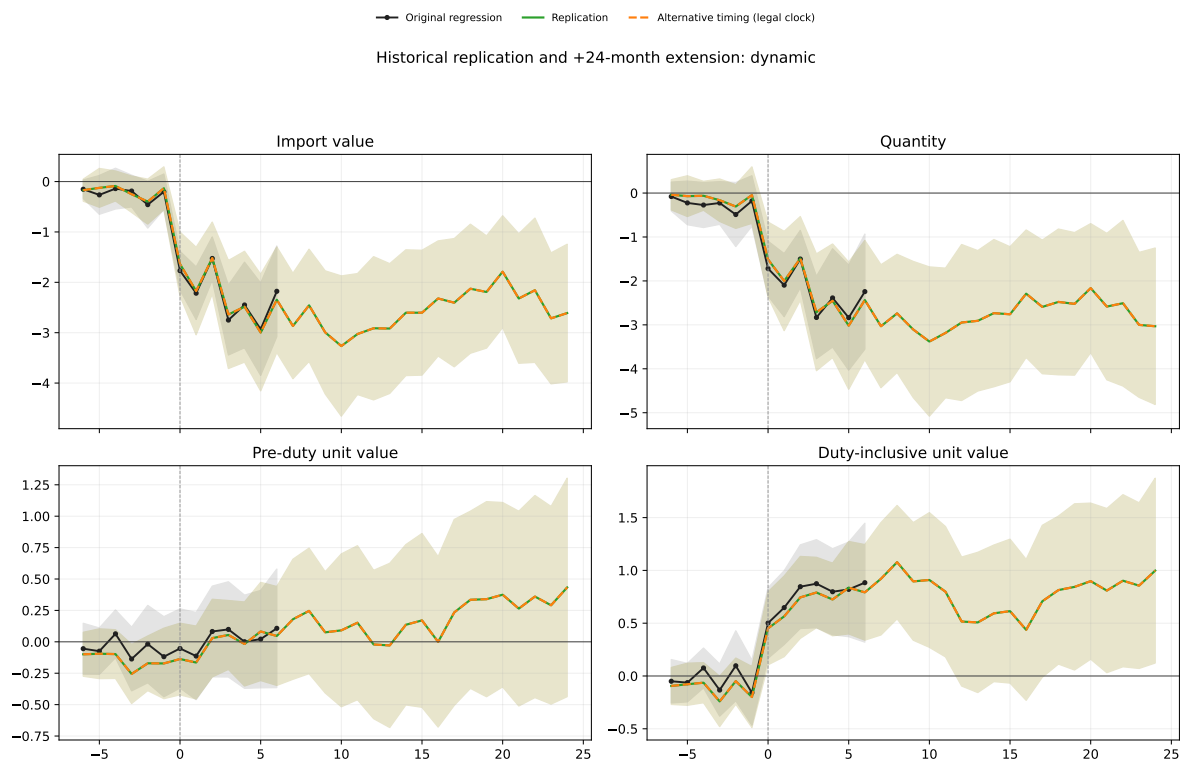


Figure 4: Historical dynamic persistence exercise, $[-6, +24]$.